



## Does poverty fuel terrorism?

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### ABSTRACT

Popular belief suggests that poverty is the primary cause of terrorism. However, several prominent studies have found little or no empirical support for this notion. I argue that these studies often overlook specific terrorist groups that recruit individuals from economically disadvantaged backgrounds. I propose conceptual frameworks that connect poor economic conditions to four distinct political objectives of terrorist groups: ethnonationalist, left-wing, right-wing, and religious fundamentalist. To support my argument, I analyze a sample of 726 terrorist groups from 1970 to 2016. My findings indicate that poverty and inequality have a significant and positive effect on the activities of left-wing and right-wing terrorist groups, but do not have the same impact on ethnonationalist and religious fundamentalist terrorist groups. Interestingly, these significant relationships diminish when conventional statistical analysis is applied to assess the effect of poverty on all terrorist activities combined. I conclude that poverty alleviation measures can be effective counterterrorism strategies, as they may ameliorate a primary motivator of economically disadvantaged individuals engaging in left-wing and right-wing terrorism.

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Conventional wisdom holds that individuals who experience frustration, anger, and hopelessness due to poverty are more likely to engage in terrorism.<sup>1</sup> For example, in 2002, President George W. Bush echoed this belief, stating, ‘We fight against poverty because hope is an answer to terror.’ He emphasized the importance of strengthening national economies, arguing that people with higher incomes are less likely to resort to terrorist violence to address their grievances. To win the War on Terror, he pledged a \$5 billion increase in aid to developing countries for innovative political and economic projects. However, empirical studies have not provided conclusive evidence linking poverty to terrorism. One main reason for this lack of evidence is that these studies often treat terrorist activities as homogeneous, overlooking that not all terrorist actions stem from economic grievances: terrorism can also be driven by ethnonationalist, religious, or other ideological factors. To better understand the connection between grievances and terrorism, it is crucial to establish a causal relationship that accounts for the diverse ideological objectives of terrorist groups (e.g. seeking economic equality).<sup>2</sup>

Ethnonationalist terrorist groups, which consist of members from specific ethnic backgrounds, aim for independence, territorial control, or autonomy from the existing

government. Left-wing terrorist groups encompass a range of ideologies, including Marxist-Leninist, Maoist, environmentalist, animal rights, anarchist, and anti-globalization movements.

In this study, I categorize terrorist groups based on their primary political ideology. Although some groups may have a mix of motives – such as left-wing terrorist groups that include members seeking to create an ethnically homogeneous country – each group usually has one dominant political goal. This categorization allows for a more structured examination of the root causes of contemporary terrorism and is a conventional method for exploring whether economic grievances specifically lead to certain types of terrorist activity (Cronin 2002-3).<sup>3</sup> For example, Robison, Crenshaw, and Jenkins (2006) classify Islamist and leftist terrorist attacks according to group ideology, stating, ‘We coded all ITERATE groups in terms of their primary ideology.’ Similarly, Hou, Gaibullov, and Sandler (2020) identify four categories of terrorist group activity based on central ideology. Other studies, such as those by Asal and Rethemeyer (2008), Blomberg, Gaibullov, and Sandler (2011), Choi (forthcoming), and Jones and Libicki (2008), also support this approach.

I propose a theoretical argument: poverty creates a sense of relative deprivation, which increases the likelihood that individuals will participate in left-wing and right-wing terrorism, while decreasing the likelihood of their engagement in ethnonationalist and religious fundamentalist terrorism. To test this argument, I conduct a quantitative analysis using a sample of 726 terrorist groups spanning the period from 1970 to 2016. I evaluate poor economic conditions in terms of economic inequality (the divide between the ‘haves’ and ‘have-nots’) and economic development (the ‘winners’ and ‘losers’ of development). Additionally, I control for various opportunity variables, such as education levels, elevation, land area in tropical regions, and land-locked status. The results show a significant negative relationship between poverty and inequality and activities associated with ethnonationalist or religious fundamentalist terrorism. In contrast, there is a significant positive relationship between poverty and inequality and activities linked to left-wing or right-wing terrorism. These findings differ from previous studies, as the effects emerge only when poverty is examined in conjunction with each of the four types of terrorist groups. Notably, these significant relationships disappear when evaluating the effect of poverty on all terrorist activities combined using conventional methods (see Table A1). This additional analysis supports my argument that previous studies mistakenly treated heterogeneous terrorist groups as homogeneous.

This study is organized into five sections. First, I present a literature review. Second, I conceptualize the relationships between poor economic conditions and the four types of terrorist group activities through the lens of relative deprivation. Third, I outline the research design. Fourth, I discuss the empirical results. Finally, I summarize the main findings and discuss the policy implications.

## Literature review

Previous studies examining the impact of economic conditions on terrorist incidents have produced inconsistent findings. Some research suggests that poverty does not

directly lead to increased terrorist activity (e.g. Campos and Gassebner 2013; Krueger and Malečková 2003). According to Gaibullov and Sandler (2019, 321), ‘to date, the literature finds no evidence of a relationship between terrorism and globalization. A similar observation holds for poverty.’ For instance, Piazza (2006) found no significant effects of poverty, malnutrition, inequality, unemployment, inflation, or poor economic growth on terrorist incidents and casualties. Additionally, Boylan (2010) demonstrates that poorer countries are not more likely to experience domestic terrorist attacks than wealthier ones.

However, a meta-analysis by Mama, Aluko, and Oumar (2022) concludes that ‘poverty influences terrorism positively.’ In a study focusing on Western European countries, Caruso and Schneider (2011) suggest a positive relationship between GDP and terrorist attacks (see also Choi 2014; Choi and Luo 2013; Plümper and Neumayer 2010). Interestingly, Shaheen et al. (2017) report mixed results, noting no significant relationship between poverty and terrorism, but a positive relationship between economic growth and terrorism. Some studies suggest a nonlinear relationship between per capita income and terrorist events (e.g. Enders, Hoover, and Sandler 2016; Jetter, Mahmood, and Stadelmann 2024).<sup>4</sup>

The inconsistent evidence and mixed findings may be due to three factors: (a) the tendency to focus on individual terrorist incidents while neglecting the activities of terrorist groups, (b) the lack of differentiation among different terrorist ideologies, and (c) the disconnect between poverty and the specific ideology of a terrorist group.

First, many studies analyze individual terrorist incidents rather than the terrorist groups that carry out these attacks: individuals with economic grievances often join terrorist activities under the guise of a terrorist group (e.g. Krueger 2007). Committing an act of terrorism requires specific training and logistical resources, such as weapons, transportation, and financing. For economically disadvantaged individuals, accessing these resources can be extremely challenging, making it nearly impossible for them to carry out terrorist attacks independently. Consequently, if they seek retribution, their best option for obtaining the necessary material and financial resources may be to either organize or join a terrorist group.

Second, treating all individual terrorist attacks as the same overlooks the fact that not all attacks arise from the same grievances. Before deciding which group to join, individuals who feel aggrieved consider whether the terrorist group shares their collective vision (Brockhoff, Krieger, and Meierrieks 2016; Drake 1998). Typically, impoverished people – often peasants and workers – are more inclined to choose left-wing terrorist groups that aim to address income inequality between the rich and poor. Those facing religious persecution are more likely to affiliate with religious fundamentalist rather than left-wing groups. Right-wing terrorist groups are generally unwellcoming to political opponents, such as radical leftists. Additionally, individuals who experience ethnic discrimination may be driven to form an ethnonationalist terrorist group to combat this discrimination, striving to protect their ethnic identity from a repressive regime. This suggests that, as different types of terrorist groups emerge in waves (Rapoport 2004), the root motivations of their members also vary. This highlights the importance of acknowledging the diversity of political motivations and objectives.

Third, existing studies that claim poverty is the root cause of all terrorist incidents often rely on a single theoretical framework (e.g. Gurr 1970; Krieger and Meierrieks 2011; Moghaddam 2005). However, terrorist motives are not uniform. While economic grievances may drive some individuals, others may be motivated by non-economic factors. A single theory cannot effectively explain the relationship between economic conditions and the diverse range of terrorist incidents.

Several scholars underscore the significance of group ideologies in understanding terrorism. Crenshaw (1981, 15) points out that ‘terrorist ideology, no matter how unrealistic, must be taken seriously as a guide to intentions.’ Similarly, Drake (1998, 78) asserts that ‘it is ideology which provides terrorists with the moral and political vision that inspires their violence’ (see also Kattelman and Burns 2023; Olzak 2022). To my knowledge, Jetter, Mahmood, and Stadelmann’s (2024, 526-527) work is the only empirical analysis that considers potential variations based on perpetrator ideology. My study differs from theirs in four key ways.

First, while Jetter, Mahmood, and Stadelmann examine the ideologies of terrorist groups in their robustness tests without providing a theoretical argument, I offer a comprehensive conceptual framework for each group’s ideology. Second, whereas Jetter, Mahmood, and Stadelmann focus on terrorism at the subnational level – looking at subregions such as Balochistan, California, Catalonia, and Île-de-France – I analyze all terrorist groups across multiple countries over the past four decades. As a result, my sample size is considerably larger than theirs. This comprehensiveness allows my findings to be more generalizable across countries and years, while Jetter, Mahmood, and Stadelmann’s conclusions are limited to specific subnational contexts. Third, while Jetter, Mahmood, and Stadelmann identify an inverted U-shaped relationship between GDP per capita and terrorism,<sup>5</sup> my goal is to uncover a linear relationship between poverty and terrorism. Finally, although they argue that the relationship between GDP per capita and terrorism remains consistent across various terrorist ideologies, I provide evidence suggesting that the link between poverty and terrorism varies depending on the ideology of the terrorist group.

## **Economic grievance-based versus political grievance-based terrorist groups**

When individuals perceive a significant gap between their expected economic status and their actual financial situation, they experience a feeling known as relative deprivation. This feeling can make them more susceptible to radical ideas and increase their likelihood of resorting to terrorist violence to address their economic grievances (Freytag et al. 2011; Gurr 1970; Iheonu and Ichoku 2021; Krieger and Meierrieks 2011; Moghaddam 2005). Relative deprivation is particularly intense among those living below the poverty line, who often feel that economic opportunities are systematically denied to them by the wealthy and powerful. They may perceive a complete lack of upward social mobility, which can foster resentment toward more affluent individuals and the influential elite. For many living in poverty, their economic struggles are viewed as a form of social injustice, leading them to consider terrorism a legitimate means of seeking redress (Niblock 2001). Countries with high inequality and limited economic mobility provide terrorist groups with a large pool of potential recruits, enhancing their capacity to carry out attacks.

However, individuals with economic grievances tend to be selective about the terrorist groups they join. They are more likely to align themselves with groups that address their economic resentments. In contrast, when individuals harbor ideological, ethnic, or religious grievances, such as those related to self-determination or religious extremism, they tend to align with non-economic groups. In other words, a terrorist group focused on economic issues can leverage economic hardships to recruit members from impoverished communities, while groups with non-economic or ideological agendas are more likely to attract individuals motivated by political or religious causes.

In short, economic grievances are not the sole motivators driving individuals toward terrorist violence and should not be regarded as the exclusive cause of all forms of terrorist activities. Relative deprivation creates an environment where both left- and right-wing terrorist groups can recruit fighters with economic grievances. However, relative deprivation does not primarily drive ethnonationalist terrorist groups, which aim to address perceived ethnic discrimination or oppression from dominant state powers, nor does it drive religious fundamentalist groups pursuing religious objectives.<sup>6</sup> Therefore, I propose categorizing terrorist group activities into two distinct types: (a) economic grievance mobilization and (b) political grievance mobilization.

### ***Economic grievance mobilization: poor economic conditions and left-wing terrorist group activity***

Left-wing terrorist groups often gain support from individuals who have economic grievances; typically, these groups seek to abolish an inequitable capitalist system. This support bolsters their recruitment efforts and facilitates the execution of their attack plans.

Such groups typically arise from poor economic conditions because their Marxist-Leninist-Maoist ideologies resonate with people from lower socioeconomic classes. They challenge the authority of the wealthy and powerful, arguing that farmers and factory workers receive far less economic benefit than they deserve. By highlighting the unequal distribution of wealth, income, and economic opportunities, they aim to garner support to replace the capitalist system with a socialist state through a proletarian revolution. Research by Kis-Katos, Liebert, and Schulze (2014) suggests that the more unequal the income distribution in a country, the more likely it is that impoverished individuals will resort to terrorism as a means of addressing these disparities. Since left-wing movements advocate for class-related demands for economic equality and egalitarianism, those living in poverty will turn to left-wing terrorist groups.

During the twentieth century, Marxism, Leninism, and Maoism emerged as the dominant political ideologies among left-wing terrorist groups. These ideologies were employed to mobilize collective forces on behalf of the poor. Following the Bolshevik Revolution, the Union of Soviet Socialist Republics (USSR) actively promoted Marxist-Leninist doctrines, leading to an increase in left-wing terrorist groups during the Cold War. However, with the collapse of the USSR in 1989, the operations of these groups declined significantly. Nevertheless, not all left-wing terrorist groups disbanded with the fall of the Soviet bloc in 1989-1990.

One notable example is the Naxalites, a left-wing terrorist group that originated from the Naxalite peasant revolt in India in 1967.<sup>7</sup> Their primary issue was land redistribution

for the landless, which galvanized poor peasants, as reflected in their slogan, ‘Land to the tiller.’ The Naxalites emerged from the split in the Communist Party of India (Marxist) in 1967, which led to several smaller factions. Even before the establishment of the Communist Party of India, Indian peasants and lower-class tribal communities had attempted to overthrow the government, which was predominantly composed of the upper classes (Sen 1982). Currently, the Communist Party of India (Marxist – Leninist) holds influence in parts of central and eastern India, highlighting the economic hardships faced by peasants and the increasing left-wing terrorist activity supported by lower-class tribal people.

Another example is the Shining Path (SL), a militant Maoist organization in Peru that aims to replace Peruvian institutions with a communist peasant revolutionary regime (Palmer 1994). Another example is the Communist Party of Nepal-Maoist (CPN-M), whose motto is, ‘Let us march ahead on the path of struggle toward establishing the people’s rule by dismantling the reactionary ruling system of the state.’<sup>8</sup> In Colombia, the Ejército de Liberación Nacional (ELN) promotes a composite communist doctrine that combines Marxism-Leninism with liberation theology to address class inequalities (Avilés 2012).

The above discussion leads to the following hypothesis:

**H1:** Poor economic conditions are more likely to increase left-wing terrorist group activity.

### ***Political grievance mobilization: poor economic conditions and ethnonationalist terrorist group activity***

Ethnonationalist terrorist groups primarily focus on issues related to ethnic identity, territory, and political rights, rather than immediate economic concerns. While relative deprivation resulting from economic discrimination against minorities can contribute to terrorism, the main motivations of these groups are non-monetary. Consequently, they tend to be less appealing to the poor and working-class individuals, who pursue economic justice.

Contrary to the belief that poverty lowers opportunity costs, participating in violent political activities often increases economic risks for many poor individuals who may consider joining ethnonationalist movements. Risks such as arrest, injury, or death can threaten the survival of their families. Additionally, the time spent training with groups focused on long-term political goals may interfere with an individual’s ability to earn a stable income, particularly since these groups often do not provide reliable wages (Hertog et al. 2024). In such circumstances, individuals generally prioritize their basic economic needs over non-essential pursuits, such as achieving ethnic autonomy or independence, which does not substantially improve their economic conditions (Hertog et al. 2024; Lee 2011).

Ethnonationalist terrorist groups engage in political movements that require a minimum level of interest, expertise, commitment, and effort, attributes more likely to be found among those with sufficient education and wealth to be concerned with issues beyond their daily economic survival. The primary tactic of these groups is to coerce the state through selective violence against civilians. To implement this tactic, they need to recruit members who are disciplined, ideologically coherent, and capable

of pursuing long-term strategic goals during extended clandestine campaigns. However, they face significant challenges in finding such individuals in impoverished communities, where people have limited educational opportunities and are preoccupied with basic economic subsistence. Consequently, these individuals are less likely to have the time or energy to engage in the organizational and ideological aspects of an ethnonationalist movement. As a result, ethnonationalist groups often struggle to maintain the necessary control for effective terrorism and may fragment into other forms of armed struggle, such as militias or insurgent groups (Abrahms 2008; Bueno de Mesquita 2005; Crenshaw 1981).

Ethnonationalist terrorist groups typically rely on volunteers who are willing to sacrifice their time and lives for the cause of reclaiming their ethnic identity. Their members often include semi-professional cadres who become aware of their marginalized ethnic status after experiencing discrimination, as well as individuals connected through ethnonationalist support networks (Epstein 2017; Tse 1998). Empirical studies support this reasoning, indicating that terrorist operatives are often better educated and more economically stable than the surrounding population (e.g. the Israeli Jewish underground). Evidence suggests that poverty actually discourages, rather than encourages, participation in ethnonationalist terrorist groups (Benmelech, Berrebi, and Klor 2012; Krueger and Malečková 2003; Piazza 2006; Sageman 2004).

The political objectives of ethnonationalist terrorist groups do not resonate with impoverished individuals; instead, they appeal to those who are ethnically awakened. While economically deprived individuals seek freedom from poverty, ethnically awakened individuals aim to liberate themselves from oppressive regimes that deny them political rights and access to public resources (Wimmer 2013). Oppressive governments often promote an 'us versus them' mentality, victimizing ethnic others (Asal and Rethemeyer 2008; Tajfel 1982). This dynamic exacerbates political animosity between dominant ethnic groups and their ethnic minorities, especially when existing sources of hostility and memories of past atrocities are at play. In response, members of ethnic minority groups are more likely to engage in ethnonationalist terrorist activities as they strive for autonomy or independence to escape ethnic persecution (Choi 2022, 2025; Horowitz 2001).

For example, when ethnic Albanians took up arms against President Slobodan Milošević's Serbian nationalism (Gagnon 2006; Toft 2012), their rebellion was a direct response to his discriminatory policies against Kosovars. Milošević aimed to reclaim Kosovo, a region of historical significance to Serbian Orthodox Christians, from its Muslim Albanian population. He accused the ethnic Albanian majority of intimidating Serbian minorities and implemented a series of laws that revoked Kosovo's provincial autonomy and dissolved its assembly. Throughout the 1990s, the Serbianized Kosovo government systematically dismissed ethnic Albanians from positions in hospitals, state-run enterprises, the judiciary, the police, and media outlets (Jenne 2009; Walasek 2015). Milošević's divisive politics posed a significant threat to the survival of ethnic Albanians. In response, they formed the Kosovo Liberation Army (KLA) to fight for Kosovo's independence from Serbia. Although economic difficulties played a role in the conflict, the primary cause was the ethnic discrimination enforced by Milošević, which led to a rise in support for the KLA (Gagnon 2006; Perritt 2008; Toft 2012).

Nevertheless, when ethnic minorities have access to fair economic opportunities in a multicultural society, they are less likely to resort to terrorism to achieve

territorial autonomy or independence (Hou 2021). For example, ethnic minorities in the U.S. often aspire to the ‘American dream’ and pursue upward mobility and economic success. Consequently, very few Americans from diverse ethnic backgrounds express a desire to join an ethnonationalist terrorist group solely for economic reasons (Krueger 2007; Pew Research Center 2017). An exception was black Americans, who faced the institutionalized economic, political, and social disadvantages of first slavery and then Jim Crow laws (Wormser 2004). However, the majority of black Americans today are not inclined to resort to terrorism to address economic grievances.

The above discussion leads to the following hypothesis:

**H2:** Poor economic conditions are less likely to lead to an increase in the activity of ethnonationalist terrorist groups.

### ***Economic grievance mobilization: poor economic conditions and right-wing terrorist group activity***

Right-wing terrorist groups, such as German neo-Nazis and the American Ku Klux Klan, often thrive in poor economic conditions by exploiting the majority group’s fear of losing economic status to minority groups.<sup>9</sup> These groups assert that their economic well-being has diminished as minority groups advance. They contend that, due to the growing economic power of these minorities, they no longer benefit from economic development, leaving them marginalized and economically threatened.

These right-wing terrorist groups frequently connect their economic anxieties to the rise of other ethnic communities, religious denominations, and social classes they deem undesirable or inferior. They often wage private wars against minority groups, including Jews, Arabs, Catholics, blacks, communists, foreign workers, and others perceived to be more prosperous than they deserve. Many right-wing terrorist groups are rooted in racism or ethnocentrism – the belief that race is an essential organizing principle in society and that certain minority groups are inherently inferior and evil (see Byman 1998; Sprinzak 1995).

As noted by Lubbers, Gijsberts, and Gijsberts (2002), Norris (2005), and Rydgren (2007), individuals who become right-wing terrorists are often from low socio-economic backgrounds. Those with low to moderate incomes and limited education usually find themselves at a disadvantage in the later stages of a country’s economic development. The modernization losers theory suggests that economic modernization hurts former middle-class workers. According to this theory, people adversely affected by economic and social transformations in society are more susceptible to the political ideologies propagated by right-wing terrorist groups (Betz 1994; Minkenberg 2003). Rapid economic development often entails significant transitions that foster resentment among individuals who become increasingly economically strained. As a result, these individuals are more likely to support right-wing terrorist groups (Kis-Katos, Liebert, and Schulze 2014). Former middle-class and blue-collar workers who once enjoyed prosperous lives now feel their share of the economic pie is shrinking.

Consequently, they scapegoat minority groups for supposedly taking over their neighborhoods and workplaces.

A notable example can be seen in the United States, where the American radical right has long been composed mainly of impoverished blue-collar and poorly educated individuals (Chermak and Gruenewald 2015).<sup>10</sup> After the defeat of the Confederacy in the Civil War, white supremacists in the South sought to reassert their influence. In 1865, they formed the Ku Klux Klan (KKK) to prevent black Americans from achieving political and economic equality. The Klan operated through small groups of masked and hooded night raiders who targeted black individuals suspected of violating the values and status of white men. These founding members of the KKK felt resentful due to economic decline, competition with free, skilled black labor, and the damage to their social status that came with it. Although initially suppressed by Union troops, the KKK reorganized and continued to promote its anti-black agenda, which expanded to include anti-Semitism, anti-Catholicism, and anti-Communism. The Vietnam War, the perceived threat of communism, and the increase in undocumented immigration further fueled the rise of more militant factions of the Klan during this period.

Scholars have noted that economic downturns significantly affect the expansion of KKK activities. The KKK often scapegoats ethnic minorities for rising white unemployment during recessions. For example, Wellman (1993) argues that blue-collar whites become anxious about their job prospects when they face competition from black and immigrant workers. Rubin (1994) suggests that increasing joblessness among whites fosters the perception that they are losing ground to minorities, who are gaining greater economic opportunities and political power. The KKK exploits these perceptions to maintain and strengthen its organization. As McVeigh (1999) points out, the Klan recruits new members by claiming that black and immigrant workers threaten economic opportunities for working-class whites.

The above discussion leads to the following hypothesis:

**H3:** Poor economic conditions are more likely to increase right-wing terrorist group activity.

### ***Political grievance mobilization: poor economic conditions and religious fundamentalist terrorist group activity***

Religious fundamentalist terrorist groups rarely recruit from the extremely poor as their core membership, but they do recruit from marginalized, lower-income populations when conditions allow. Ideology, identity, and political grievance matter more than poverty alone (Berrebi 2007; Davis 2019; Krueger and Malečková 2003; Sageman 2004). Extreme poverty can undermine the ideological discipline, operational security, and recruitment efficiency on which these groups rely. Instead of seeking out individuals driven by economic desperation, these groups selectively recruit people who possess a certain level of material stability, education, technical skills, and social connections. This selection process enables recruits to commit to clandestine activities for long periods. The reason for this selectivity is that religious fundamentalist movements require strong ideological

commitment and tactical proficiency. Individuals whose basic survival needs dominate their lives may struggle to engage in high-risk activities.

Religious fundamentalist terrorist groups pursue overarching and absolute goals, such as establishing a caliphate, purifying society, and fulfilling divine commands. To achieve these religious objectives through terrorism, members must be willing to forgo material rewards, resist pressure to defect, strictly obey authority, and maintain doctrinal consistency. However, individuals living in extreme poverty are vulnerable to compromise by security forces that offer financial incentives in exchange for information. As a result, these religious groups rationally choose to recruit individuals from middle-class or wealthier backgrounds to maintain their operational integrity (Abrahms 2008; Bueno de Mesquita 2005; Crenshaw 1981).

Sageman (2004) analyzes the profiles of 173 militants associated with Al-Qaeda and finds that approximately three-quarters of global Salafi mujahedin come from upper or middle-class backgrounds. Krueger and Malečková (2003) reveal that terrorists from Hezbollah, Israeli-Jewish groups, and Palestinian suicide bombers tend to be wealthier than the general population. Additionally, Krueger (2007) shows that many suicide bombers actually originate from higher socioeconomic backgrounds. Berrebi (2007, 51) concludes that 'if there is a link between income level, education, and participation in terrorist activities, it is either very weak or in the opposite direction of what one intuitively might have expected.' Furthermore, Choi (2015) finds that income inequality does not significantly affect suicide terrorism carried out by religious fundamentalists, and that economic growth has only a limited impact. This body of evidence suggests that poor economic conditions are not the primary driving force behind religious fundamentalist terrorist activity.

The recruitment patterns of religious fundamentalist terrorist groups are in line with these empirical findings. They recruit new members through advanced religious education, including madrasas and seminaries, university-based religious movements, urban mosques, clerical mentorship, and transnational ideological networks. While individuals from poor backgrounds may attend these educational institutions, they are generally less likely to gain access to higher levels of religious or secular education, participate in elite ideological circles, or develop connections with influential religious authorities. As a result, poverty limits access to recruitment pathways within religious fundamentalist groups, even in highly religious societies (Della Porta 2013; Horgan 2008; Wiktorowicz 2005).

Religious fundamentalist terrorist groups are more likely to act on the belief that terrorist violence is a legitimate means to protect and preserve their religious identity. They often use religious scriptures to justify their violent acts against other belief systems and to recruit new members (Cromartie 2005). The denial of their religious beliefs by non-believers is perceived as a violation of their religious sanctity. For example, groups like Al-Qaeda, the Islamic State of Iraq and Syria, Hamas, and Hezbollah maintain that 'Islam and the West are culturally incompatible and cannot coexist within the same geographic space' (Masters 2008, 398).

In recent years, Islamic fundamentalists have developed a culture that glorifies martyrdom (Benmelech, Berrebi, and Klor 2012; Krueger 2007). As a result, they have adopted suicide bombings as one of their most effective tactics to retaliate against those who suppress their religious beliefs. Indeed, religious fundamentalists are often more willing to

die for their causes than other types of terrorists. In contrast, different types of terrorists tend to view suicide attacks as a significant loss of manpower and popular support, rarely using this tactic.<sup>11</sup> For religious fundamentalists, suicide bombings do not improve their personal economic situations since they will perish without receiving any monetary reward for themselves, and only occasionally for their family members. This indicates that their willingness to sacrifice their lives is driven more by their religious conviction than by a desire for personal economic betterment.

The above discussion leads to the following hypothesis:

**H4:** Poor economic conditions are less likely to lead to an increase in the activity of religious fundamentalist terrorist groups.

## Research design

To empirically test the four hypotheses outlined above, I utilize a sample of 726 terrorist groups spanning 101 countries from 1970 to 2016.<sup>12</sup> This sample is consistent with the data collection by Hou, Gaibullov, and Sandler (2020) regarding terrorist groups, countries, and years. Consequently, my sample inherits both the strengths and weaknesses of the original data, including the definitions of terrorism and terrorist groups. To avoid data issues, the original researchers – Hou, Gaibullov, and Sandler – excluded countries with little to no likelihood of terrorist attacks. This aligns with standard practices in the field of terrorism and conflict studies.

I set the terrorist group activity-year as the unit of analysis, as my theoretical argument views terrorist group activity as a violent expression of economically deprived individuals. This approach is preferable to using individual terrorists as the unit of analysis, as individuals rarely carry out terrorist attacks independently. Most aggrieved individuals join a terrorist group, where they receive training and logistical support that enable them to engage in acts of terrorist violence. I build a baseline model as follows:

$$\text{TerroristGroupActivity}_{it} = \beta_0 + \beta_1 * \text{EconomicInequality}_{it-1} + \beta_2 * \text{EconomicDevelopment}_{it-1} + Z * \text{ControlVariables}_{it-1} + \varepsilon_{it}$$

where *Terrorist Group Activity* is a function of *Economic Inequality* and *Economic Development*, along with controls. *Control Variables* is a vector of controls, and *Z* is a vector of corresponding coefficients. All right-hand-side variables are lagged one year to address potential endogeneity.

## Dependent variables

I create two dependent variables: one dichotomous and the other a count variable. To ensure the robustness of empirical findings, I also employ the country-year as the secondary unit of analysis. For this unit of analysis, I create a dependent variable measured in counts.

The first dependent variable is a dichotomized measure per terrorist group activity-year. I assign '1' to ethnonationalist terrorist activity when aggrieved individuals resort to violence through a terrorist group aiming to create an ethnically homogeneous territory

in any year and '0' otherwise. For instance, all terrorist activity-years for Northern Ireland are coded as '1' since it experienced ethnonationalist terrorist violence. In the U.K., some ethnonationalist terrorist activity-years are coded as '1' (e.g. the Irish Volunteers in 2016), while other non-ethnonationalist terrorist activity-years are coded as '0' (e.g. the English Defence League in 2016). All ethnonationalist terrorist activity-years are coded as '0' for Switzerland, as the violence there was not ethnonationalist. This operationalization allows the dependent variable *in the cross-national, time-series data* to vary by terrorist group, country, and year. As mentioned earlier, when terrorist groups have multiple political ideologies, I focus on their primary goal. For instance, even though Hamas is associated with both ethnonationalist and religious ideologies, it is categorized as ethnonationalist according to its primary political objective.

I assign '1' for left-wing terrorist activity with Marxist-Leninist, Maoist, environmentalist, animal rightist, anarchical, and anti-globalist ideologies,<sup>13</sup> and '0' otherwise; '1' for right-wing terrorist activity with racist, fascist, and neo-Nazi ideologies, and '0' otherwise; and '1' for religious fundamentalist terrorist activity seeking to redress the persecution of their religious beliefs, and '0' otherwise.

Since this dependent variable is dichotomous, I use logistic regression. To ensure unbiased standard errors of logit coefficients in the presence of heteroscedasticity, I specify robust standard errors.

I gather the four types of terrorist group activity from Extended Data on Terrorist Groups (EDTG), compiled by Hou, Gaibullov, and Sandler (2020).<sup>14</sup> EDTG has several advantages, including being the most up-to-date dataset at the group level and systematically excluding criminal and drug trafficking groups (see Hou 2021 for more details). After cleaning EDTG,<sup>15</sup> I identified 726 terrorist groups over the period from 1970 to 2016. Among these, I find 296 ethnonationalist terrorist activities (41 percent), 192 left-wing terrorist activities (26 percent), 35 right-wing terrorist activities (5 percent), and 203 religious fundamentalist terrorist activities (28 percent). Because the frequency of each type exceeds one percent, no terrorist group activity is considered rare.

The second dependent variable is the total number of terrorist events per terrorist group activity-year. For example, in 1970, the US had seven active terrorist groups subscribing to leftist ideologies. I record '2' terrorist events for the Revolutionary Action Party, '5' for Revolutionary Force Seven, '14' for the Weather Underground, and '0' for the remaining four non-leftist groups. Since this dependent variable consists of counts and has a variance significantly greater than the mean, I use negative binomial regression.

The third dependent variable captures the total number of terrorist group activities per country-year, counting those associated with each of the four ideologies in a given year. Because this dependent variable is also a count, and its variance is not substantially larger than the mean, I employ Poisson regression as the estimation method.

### **Main independent variables**

Consistent with my conceptualization of poor economic conditions, I employ two indicators: economic inequality and economic development. These indicators are widely used, although there are other ways to assess economic hardship (for a discussion on the strengths and weaknesses of economic indicators, see Hertog et al. 2024; Mama, Aluko, and Oumar 2022, 53-55).

Economic inequality refers to the unequal distribution of income and opportunities among different social groups, effectively highlighting the conditions that lead to relative deprivation. I measure economic inequality using the Gini coefficient, drawing on data from UNU-WIDER's (2017) World Income Inequality Database (WIID3.4).<sup>16</sup> In contrast, economic development signifies an increase in an economy's productive capacity, resulting in greater output of goods and services. While economic development is generally linked to improved living standards, it can also create disparities, leading to heightened feelings of relative deprivation. I quantify economic development as the logarithm of real gross domestic product (GDP) per capita in constant 2010 US dollars, using data from the World Bank's (2018) World Development Indicators (WDI) data collection. Since GDP per capita and its changes yield similar estimates, and the former has been widely used in previous studies, I focus on it in the Empirical Results section.

### ***Control variables***

To avoid producing spurious estimates, I include control variables known to be related to terrorist events. These variables consist of education levels, democracy and its squared term, ethnic fractionalization and its squared term, international conflict, internal conflict, elevation, land area in tropical regions, whether a country is landlocked, and seven regional variables.

Individuals with lower levels of education are more likely to join terrorist groups as foot soldiers, while those who are highly educated often assume organizational and leadership roles, leveraging their extensive knowledge, experience, and skills (Benmelech, Berrebi, and Klor 2012). To measure education, I use the average years of education among citizens aged 15 and older, drawing on Clio Infra data.<sup>17</sup>

The political environment, particularly the presence of democratic institutions and ethnic diversity, can significantly influence the level of terrorist activity. The relationship between democracy and terrorism is not straightforward. While increased democratization is initially associated with lower levels of terrorism, it can reach a threshold beyond which it may become a liability, increasing terrorist activity (Gaibullov, Piazza, and Sandler 2017). To capture this dynamic, I include both democracy and its square term, utilizing data from the Polity IV project (Marshall and Jaggers 2017). Using the same reasoning as the democracy variables, I incorporate an index of ethnic fractionalization and its squared term, based on data from Alesina et al. (2003).

To measure the security environment faced by terrorist groups, I include two variables: international conflict and internal conflict (Choi and James 2017; James and Choi 2018). These measures are dichotomous, coded as '1' if a conflict occurred and '0' otherwise. This data comes from Brecke's (2001) Conflict Catalog. I incorporate three geographical features of each country: the logarithm of the country's average elevation, the percentage of land territory located in the tropics, and a dummy variable coded as '1' if the country is landlocked (Hou, Gaibullov, and Sandler 2020). To account for the base country's geographic location, I create seven regional dummy variables.

Table 1 presents the descriptive statistics for all variables utilized in the study.

**Table 1.** Descriptive statistics.

| Variable                                          | Obs   | Mean    | Std. Dev. | Min    | Max    |
|---------------------------------------------------|-------|---------|-----------|--------|--------|
| Nationalist Terrorist Group Activity              | 9,098 | 0.466   | 0.499     | 0      | 1      |
| Left-Wing Terrorist Group Activity                | 9,098 | 0.244   | 0.429     | 0      | 1      |
| Right-Wing Terrorist Group Activity               | 9,098 | 0.029   | 0.169     | 0      | 1      |
| Religious Fundamentalist Terrorist Group Activity | 9,098 | 0.261   | 0.439     | 0      | 1      |
| Income Inequality                                 | 7,004 | 38.393  | 7.929     | 20.100 | 69.500 |
| Economic Development                              | 8,573 | 7.998   | 1.507     | 5.098  | 11.236 |
| Education                                         | 7,462 | 5.947   | 3.080     | 0.171  | 13.482 |
| Democracy                                         | 8,854 | 13.592  | 6.041     | 0      | 20     |
| Democracy <sup>2</sup>                            | 8,854 | 221.244 | 139.259   | 0      | 400    |
| Ethnic Fractionalization                          | 9,037 | 0.449   | 0.222     | 0.012  | 0.902  |
| Ethnic Fractionalization <sup>2</sup>             | 9,037 | 0.251   | 0.209     | 0.000  | 0.814  |
| International Conflict                            | 7,094 | 0.085   | 0.279     | 0      | 1      |
| Internal Conflict                                 | 7,151 | 0.262   | 0.440     | 0      | 1      |
| Elevation                                         | 9,065 | 6.387   | 0.631     | 3.405  | 7.850  |
| Land Territory in the Tropics                     | 9,065 | 0.378   | 0.417     | 0      | 1      |
| Landlocked                                        | 9,098 | 0.070   | 0.256     | 0      | 1      |
| Eastern Asia and Pacific                          | 9,098 | 0.133   | 0.340     | 0      | 1      |
| Europe and Central Asia                           | 9,098 | 0.181   | 0.385     | 0      | 1      |
| Latin America and Caribbean                       | 9,098 | 0.085   | 0.279     | 0      | 1      |
| Middle East and North Africa                      | 9,098 | 0.218   | 0.413     | 0      | 1      |
| North America                                     | 9,098 | 0.035   | 0.183     | 0      | 1      |
| South Asia                                        | 9,098 | 0.295   | 0.456     | 0      | 1      |
| Sub-Saharan Africa                                | 9,098 | 0.133   | 0.340     | 0      | 1      |

## Empirical results

Table 2 presents logit estimates indicating that the effects of poor economic conditions vary across terrorist groups' ideologies. As hypothesized, relative economic disparity is more likely to drive people toward both left-wing and right-wing terrorism. However, I find that individuals who face ethnic and religious persecution are less inclined to engage in terrorism. This suggests that their motivation to resort to terrorism is primarily rooted in their experiences of ethnic and religious persecution, rather than economic distress.

In Model 1, *Economic Inequality* exhibits a statistically significant negative relationship, indicating that a more unequal distribution of income and opportunities is associated with a lower likelihood of involvement in ethnonationalist terrorist activities. This finding supports my hypothesis: Poor economic conditions are less likely to lead to increased activity by ethnonationalist terrorist groups. Put differently, ethnic minorities tend to demand political rights only after achieving a reasonable level of economic stability. Economic security and improved well-being provide the foundation for aspirations toward personal autonomy, participation, and influence in governance. Once ethnic others have their basic needs met and attain a degree of economic independence, they are better positioned to seek broader freedoms, express their concerns, and advocate for systems that safeguard their gains and foster further social progress. As ethnic minorities become more economically affluent while being systematically denied the political and cultural power enjoyed by the dominant ethnic group, their demand for inclusion and equality intensifies. This growing frustration can lead to the emergence of liberation movements, including those by ethnonationalist terrorist groups.

As hypothesized, Model 2 reveals that *Economic Inequality* is positively associated with left-wing terrorist activity. When impoverished individuals face limited opportunities for



upward social mobility and feel a sense of relative deprivation, they are more likely to engage in left-wing terrorism in their pursuit of economic justice. In Model 3, for right-wing terrorist activity, *Economic Inequality* is also statistically significant and aligns with my expectations. Individuals who are concerned about their declining economic status are more inclined to blame minority groups and participate in right-wing terrorist activities.

*Economic Inequality* in Model 4 is negatively associated with religious fundamentalist terrorist activity. Economic hardships do not serve as the main motivator for these actions. This finding supports the notion that economic struggles do not directly correlate with religious fundamentalist terrorism. In other words, religious fundamentalist terrorist groups become more active when they recruit individuals who, despite experiencing better economic conditions, still feel alienated and unfairly treated because of their religious beliefs. This sense of injustice arises because their rights to religious freedom have not improved as much as their economic rights.<sup>18</sup>

In Models 5 to 8, I examine another measure of poor economic conditions, *Economic Development*, for each of the four terrorist groups. The analysis of the association between economic development and terrorism in Models 5-8 reveals empirical regularities similar to those found in Models 1-4. In these models, *Economic Development* has a significant negative impact on both ethnonationalist and religious fundamentalist terrorist activities and is a significant positive predictor of left-wing and right-wing terrorist activities.

Jetter, Mahmood, and Stadelmann (2024) examine terrorist activity at the subnational level using a small sample. They report an inverted U-shape relationship between GDP per capita and various ideological motivations for terrorism (see also Enders, Hoover, and Sandler 2016). While my study does not explicitly address non-linearity, I conducted preliminary tests and found no nonlinear relationship between GDP per capita and right-wing terrorism. I discovered a U-shaped relationship between GDP per capita and ethnonationalist terrorism. However, I found an inverted U-shape relationship between GDP per capita and both left-wing terrorism and religious fundamentalist terrorism.<sup>19</sup> These results differ from those of Jetter, Mahmood, and Stadelmann because my sample includes terrorist activities from all groups within nation-states and spans multiple countries.

To examine which economic condition better explains the four types of terrorist activity, I include both *Economic Inequality* and *Economic Development* in the same model. Comparing the relative importance of the two economic conditions requires standardizing the coefficients. I display the results in Models 9-12 in Table 2. The results indicate that the relative importance varies across terrorist groups' ideologies. For left-wing terrorist activity, *Economic Inequality* is found to be more significant than *Economic Development*. This relationship, however, reverses for right-wing terrorist activity. These findings suggest that impoverished individuals, such as peasants and factory workers, may view left-wing terrorist groups as a more effective means of expressing their class resentment, anger, and deprivation compared to other types of terrorist groups. Additionally, those who are dissatisfied with the shifting socio-economic status of their racial or social group and believe they deserve greater economic benefits than minority groups are more likely to engage in right-wing terrorist activity.

The relative importance of the two economic factors varies, as demonstrated in Models 9-12 in Table 2. However, the overall pattern across the four types of terrorist activities remains consistent in terms of the coefficients and their

statistical significance. Specifically, two types of terrorist activities have positive correlations with economic conditions, while the other two show negative associations (see Table A3). Ethnonationalist and religious fundamentalist terrorist activities are less likely to occur during periods of economic distress. In contrast, left-wing and right-wing terrorist activities tend to increase when people experience economic hardships.

To confirm the robustness of the above findings, I re-estimated Table A3 by changing the dependent variable from a dichotomous measure to a count measure (i.e. the second dependent variable). The results, presented in Table A4, are from negative binomial regression models similar to those in Table A3. These findings indicate that individuals in economic distress are more likely to participate in both left-wing and right-wing terrorist activities.

To further validate these findings, I reran the analysis presented in Table A3 using a different dependent variable: country-year counts. It is important to note that aggregating the data at this level results in a significantly smaller sample. The results of the Poisson regression, presented in Table A5, closely align with those from the logistic regression in Table A3.

To avoid spurious relationships between poor economic conditions and terrorist activity, I control for other factors that may influence both the outcome and the main explanatory variables. Table 3 presents the results of multivariate logistic regression models. It is important to note that Table 3 includes fewer observations than Table 2 because the control variables have fewer data points than the main independent variables. When examining the significance and direction of poor economic conditions, the overall results in Table 3 align with those in Table 2. These findings emphasize that economically deprived individuals do not indiscriminately join available terrorist groups. Instead, they are selective in choosing the appropriate organization to address their specific grievances. All other factors being equal, economic grievances show a negative association with ethnonationalist and religious fundamentalist terrorist activity (Models 13 and 16) while being positively associated with left-wing and right-wing terrorist activity (Models 14 and 15).

Interestingly, none of the control variables resembles the general pattern of poor economic conditions.<sup>20</sup> For example, *Education* emerges as a significant and positive predictor of ethnonationalist terrorist activity, a significant and negative driver of left-wing terrorist activity, an insignificant and positive trigger of right-wing terrorist activity, and a significant and negative determinant of religious fundamentalist terrorist activity. This suggests that highly educated people are more likely to experience ethnonationalist awakenings and, as a result, participate in ethnonationalist terrorist activities. However, they are less likely to engage in left-wing and religious fundamentalist terrorist activities and are not prone to commit right-wing terrorism.

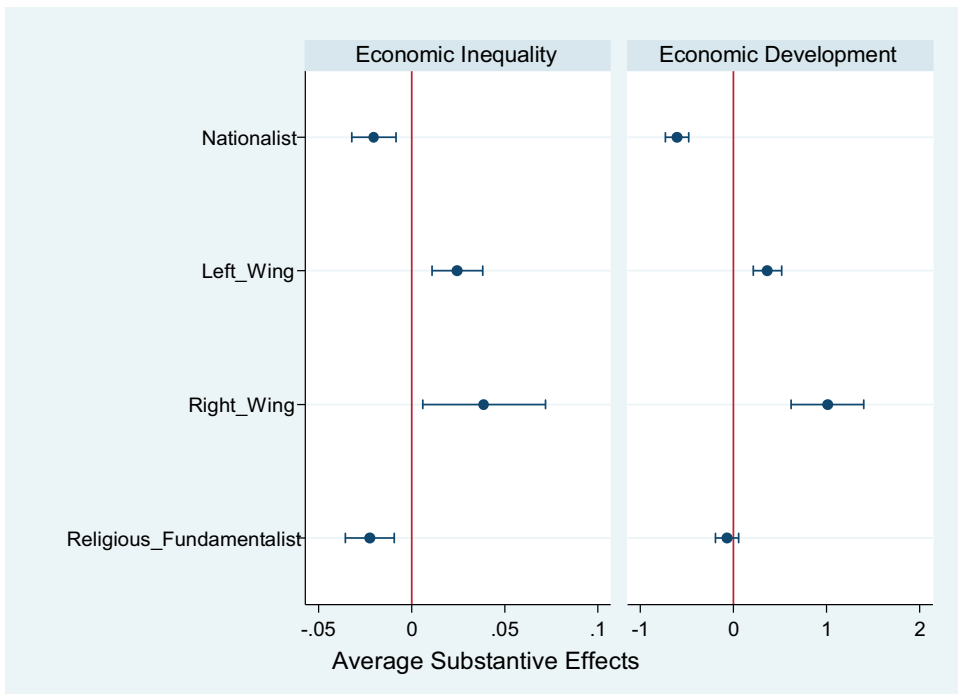
Table 3 also accounts for regional differences. Compared to the Middle East and North Africa, the other regions show statistically significant results across the four models, with varying effects depending on the ideology of the terrorist groups. For instance, ethnonationalist terrorist activity is unlikely to emerge in North America, where multiculturalism is the prevailing norm. In contrast, the other three types of terrorist groups are more likely to engage in violent actions. The inclusion of regional dummy variables is crucial, as some areas are more susceptible to terrorism than others.

**Table 3.** Economic conditions and four types of terrorist group activity: control variables added.

|                                       | Nationalist           | Left-Wing             | Right-Wing            | Religious Fundamentalist |
|---------------------------------------|-----------------------|-----------------------|-----------------------|--------------------------|
|                                       | Model 13              | Model 14              | Model 15              | Model 16                 |
| Economic Inequality                   | −0.020***<br>(0.006)  | 0.025***<br>(0.007)   | 0.039**<br>(0.017)    | −0.022***<br>(0.007)     |
| Economic Development                  | −0.606***<br>(0.063)  | 0.365***<br>(0.078)   | 1.009***<br>(0.200)   | −0.070<br>(0.065)        |
| Education                             | 0.257***<br>(0.027)   | −0.256***<br>(0.034)  | 0.092*<br>(0.055)     | −0.134***<br>(0.025)     |
| Democracy                             | −0.110***<br>(0.031)  | −0.411***<br>(0.044)  | −0.091<br>(0.134)     | 0.459***<br>(0.035)      |
| Democracy <sup>2</sup>                | 0.004***<br>(0.001)   | 0.021***<br>(0.002)   | 0.008<br>(0.005)      | −0.020***<br>(0.001)     |
| Ethnic Fractionalization              | 14.880***<br>(1.036)  | −15.619***<br>(0.960) | −3.667*<br>(1.971)    | −0.742<br>(0.824)        |
| Ethnic Fractionalization <sup>2</sup> | −17.184***<br>(1.065) | 15.325***<br>(1.030)  | 7.070***<br>(2.528)   | 2.808** *<br>(0.882)     |
| International Conflict                | 0.256**<br>(0.116)    | −0.088<br>(0.165)     | −0.612*<br>(0.357)    | −0.262**<br>(0.123)      |
| Internal Conflict                     | 0.101<br>(0.079)      | 0.262**<br>(0.101)    | −1.291***<br>(0.324)  | −0.392***<br>(0.091)     |
| Elevation                             | −1.362***<br>(0.109)  | 1.287***<br>(0.111)   | −0.081<br>(0.184)     | −0.039<br>(0.077)        |
| Land Territory in the Tropics         | −1.819***<br>(0.197)  | 1.917***<br>(0.212)   | −0.566<br>(0.394)     | −1.633***<br>(0.135)     |
| Landlocked                            | 0.927***<br>(0.214)   | 1.178***<br>(0.202)   | dropped               | −2.320***<br>(0.329)     |
| Eastern Asia and Pacific              | 0.792***<br>(0.151)   | 0.873***<br>(0.180)   | −0.978**<br>(0.473)   | 0.417***<br>(0.118)      |
| Europe and Central Asia               | 0.644***<br>(0.129)   | 1.891***<br>(0.164)   | −1.151***<br>(0.213)  | −1.180***<br>(0.127)     |
| Latin America and Caribbean           | −2.322***<br>(0.359)  | 4.549***<br>(0.278)   | 0.901**<br>(0.344)    | dropped                  |
| North America                         | −2.825***<br>(0.284)  | 2.408***<br>(0.215)   | 0.779***<br>(0.226)   | 0.454**<br>(0.211)       |
| South Asia                            | 0.953***<br>(0.143)   | 0.605***<br>(0.214)   | 1.661***<br>(0.580)   | −0.782***<br>(0.155)     |
| Sub-Saharan Africa                    | 3.902***<br>(0.297)   | −0.447<br>(0.315)     | 1.139*<br>(0.597)     | −1.275***<br>(0.217)     |
| Constant                              | 10.515***<br>(0.840)  | −9.640***<br>(0.953)  | −14.920***<br>(2.429) | 0.183<br>(0.700)         |
| Wald Chi <sup>2</sup>                 | 659.41                | 1016.37               | 548.45                | 891.23                   |
| Prob > Chi <sup>2</sup>               | 0.001                 | 0.001                 | 0.001                 | 0.001                    |
| Pseudo R <sup>2</sup>                 | 0.225                 | 0.364                 | 0.278                 | 0.138                    |
| Observations                          | 6,058                 | 6,058                 | 6,058                 | 6,058                    |

Note: Robust standard errors are in parentheses. \* is 0.10, \*\* is 0.05, and \*\*\* is 0.01, two-tailed tests.

Statistical significance alone may not always indicate practical relevance, so in [Figure 1](#), I depict the average substantive effects of two economic conditions—*Economic Inequality* and *Economic Development*—based on the odds (probability) of the presence of each type of terrorist group, as indicated in Models 13 to 16 of [Table 3](#). Average substantive effects provide insight into the average change in predicted values when either *Economic Inequality* or *Economic Development* increases by one unit. For example, the left side of the figure shows that a one-unit increase in *Economic Inequality* is, on average, expected to encourage impoverished individuals to participate in left-wing or right-wing terrorist activities, while discouraging their involvement in ethnonationalist or religious



**Figure 1.** Average substantive effects with 95% confidence intervals.

fundamentalist groups. This graphical analysis clearly indicates that certain types of terrorist activities are indeed rooted in economic grievances.

## Conclusion

Many people believe that poverty is a primary driver of increased terrorism. However, academic research has not consistently produced strong evidence to support this widespread belief. This discrepancy can be attributed to three key factors: (a) an overemphasis on individual terrorist attacks rather than on group activities, (b) the dilution of the significance of economic grievances by relying on aggregate measures of heterogeneous terrorist events, and (c) a disconnect between the poverty theory and the distinct ideologies of various terrorist groups. To address these shortcomings, I disaggregate terrorist group activities by political ideology. I find a significant and *positive* correlation between poor economic conditions and left-wing and right-wing terrorist activities, but not with other types. This suggests that impoverished individuals, such as farmers and factory workers, are more likely to engage in left-wing and right-wing terrorism as a means of addressing economic resentment and deprivation.

In contrast, I discover a significant and *negative* effect of poor economic conditions on ethnonationalist and religious fundamentalist terrorist activities. This indicates that poverty is not the underlying cause of ethnic and religious terrorism; rather, it may be a secondary outcome of discrimination based on ethnicity or religion. Those who experience such discrimination may also face economic marginalization.

Alternatively, this finding may imply that when ethnic and religious identities are the primary sources of societal conflict, they overshadow other socioeconomic issues that affect a smaller subset of disadvantaged individuals. In these instances, the armed struggles led by these groups tend to prioritize their specific identities over poverty-related grievances, as their support base predominantly consists of the targeted ethnic or religious population. Examples include the Free Papua Movement and the Provisional Irish Republican Army.

The findings of this study have important policy implications. Many prominent studies have failed to establish a connection between poverty and terrorism, leading scholars, policymakers, and journalists to dismiss the idea that addressing poverty could be an effective counterterrorism strategy. However, my empirical results indicate otherwise. It is time to revisit previous research and incorporate this new evidence. Poor economic conditions can indeed play a role, particularly depending on the ideology of the terrorist groups involved. When individuals have stable job prospects, it becomes more difficult for left- and right-wing terrorist groups to recruit them. Therefore, the counterterrorism community should implement public works employment programs for unemployed youth, expand microfinance and small-business grants in marginalized regions, and offer job training that aligns with local market needs rather than generic training programs.

## Notes

1. In this study, I adopt LaFree and Dugan's (2007, 184) definition of terrorism as 'the threatened or actual use of illegal force and violence to attain a political, economic, religious or social goal through fear, coercion or intimidation.'
2. I borrow Hou, Gaibullov, and Sandler's (2020, 201) definition of a terrorist group – 'a subnational collective whose members try to obtain a political goal by intimidating an audience (or constituency) through violence or its threat.' Right-wing terrorist groups include those that are racist, fascist, and neo-Nazi in nature. Religious fundamentalist terrorist groups engage in acts of terrorism to fulfill a religious mandate or to coerce others into adopting their beliefs.
3. This convention reduces precision since some terrorist groups may have mixed political objectives.
4. See also Abadie (2006), Berman et al. (2011), Blair et al. (2013), and Freytag et al. (2011).
5. Enders, Hoover, and Sandler (2016) conduct an innovative study exploring the ideologies of terrorist groups and argue for nonlinear shifts in the relationship between GDP per capita and terrorism.
6. Despite not being the primary driver of these groups, some individuals of course join ethnonationalist or religious fundamentalist terrorist groups due to economic grievances.
7. The term, Naxal, derives from the name of the village Naxalbari in West Bengal, India.
8. See <https://www.satp.org/satporgtp/countries/NEPAL/terroristoutfits/index.html>.
9. An alternative explanation for the rise of right-wing terrorist groups is that some individuals may join these groups due to cultural and sociological grievances. However, I do not explore this possibility further, as it is not the primary focus of this study and is challenging to measure across different countries and time periods. Another consideration is that some right-wing terrorist groups, such as the Argentine Anticommunist Alliance, emerged as a response to left-wing terrorism. Nonetheless, communists, who comprise the core of left-wing terrorist groups, represent only a small fraction of the various minority groups that are targeted by right-wing terrorists worldwide.

10. Piazza (2017), however, finds that economic hardships, such as poverty, the decline of manufacturing employment, and the 'Farm Crisis,' are not associated with right-wing terrorism.
11. Notable exceptions are the Liberation Tigers of Tamil Eelam and the Kurdistan Workers' Party.
12. Upon publication, the replication materials will be accessible at <https://dataverse.harvard.edu/>.
13. The occurrence of environmental and animal rights terrorism in the sample is relatively low compared to other left-wing terrorist activities. As a result, its influence on my statistical analysis is minimal.
14. Hou, Gaibullov, and Sandler (2020) use the four types of terrorist group activity as predictors in the same model (e.g. Table 5 on p. 219). In contrast, I employ each of these types as an outcome variable in separate models. As of January 11, 2026, Hou, Gaibullov, and Sandler's study has been cited 120 times, including Kattelman and Burns (2023) and Olzak (2022).
15. For example, I remove terrorist groups that operate in those countries that the Correlates of War Project does not assign country codes (see <https://correlatesofwar.org/data-sets>).
16. See <https://www.wider.unu.edu/database/world-income-inequality-database-wiid34>.
17. See <https://clio-infra.eu/>.
18. High-income countries may have low poverty rates, with few individuals living on very low daily incomes, but they can still exhibit significant income inequality within their borders. To address this issue, I re-run Models 1 to 4 after excluding 72 high-income countries, as classified by the World Bank in 2016 based on GNI per capita thresholds (approximately \$12,235 or higher). 2016 marks the end of my sample period. Table A2 presents the estimates obtained from this revised analysis, which do not deviate from the main findings.
19. Future research should further explore the nonlinear causality between GDP per capita and different terrorist groups.
20. As a robustness check, I included two additional controls: state capacity, defined as the government's authority over its territory in the V-Dem dataset, and the age of the group involved. Recent research suggests that state capacity is linked to terrorist activity (Gaibullov, Piazza, and Sandler 2024, 2025), and the age of a terrorist group is a significant factor in its terrorist activity (Hou, Gaibullov, and Sandler 2020). As shown in Table A6, the main findings of this study remain consistent even after accounting for these additional controls. In addition, I include country fixed-effects to control for the time-invariant characteristics of each country and year fixed-effects to account for the prevalence of terrorism that occurs in waves. To address the issues of serial correlation and unmodeled shocks within each country, I compute country-clustered standard errors. The results, presented in Table A7, are similar to those in Table 3.

## Disclosure statement

No potential conflict of interest was reported by the author(s).

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